

Yizhe Zhang

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Research Interest

My recent research focuses on pushing the LLM to gain more intuition and strong generalization, especially in the code domain, regarding: 1) code LLM/agent; 2) long-horizon planning; 3) RAG/reasoning with continuous tokens; 4) text diffusion models; 5) coding-based AI scientist.

Professional Experience

Staff Research Scientist, Apple, Cupertino, CA

05/2022 – Present

Manager: Navdeep Jaitly, Ronan Collobert

Using code domain as a testbed for planning and reasoning. Led development of:

- **Latent Reasoning: LaDiR, LaDiRL, CLaRa** – Continuous latent reasoning for LLMs
- **DiffuCoder** – Diffusion-based code generation (700 GitHub stars in one month)
- **SWE-Gym** – Code agents training framework with 1.8M+ dataset downloads per month

Research Scientist, Facebook AI, Menlo Park, CA

09/2021 – 05/2022

Manager: Yashar Mehdad

Worked on Retrieval-Augmented Generation (RAG) and Summarization systems.

Senior Researcher, Microsoft Research, Redmond, WA

02/2018 – 09/2021

Manager: Bill Dolan

Led development of generative dialogue models:

- **DialoGPT** – Open-sourced pretrained chat model with 1,912 citations, 2.4k GitHub stars

Education

Ph.D. in Computational Biology, Duke University, Durham, NC

08/2013 – 02/2018

Advisor: Prof. Lawrence Carin

Dissertation: Efficient and Scalable Markov Chain Monte Carlo Methods

Research Focus: VAE, GAN, MCMC for text generation

M.Sc. in Statistics, Duke University, Durham, NC

08/2015 – 02/2018

B.Sc. in Physics, Nanjing University, Nanjing, China

08/2007 – 06/2011

Professional Services

Area Chair: NeurIPS (since 2020), ICML (since 2022), ICLR (since 2023), ACL (2020-2021), EMNLP (2022), NAACL (2023), AAAI (2018-2021)

Action Editor: TMLR (since 2023), ARR (since 2023)

Organization Committee: ACL (2020)

Reviewer: NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AISTATS, AAAI

Recent Talks

Towards Understanding and Building Intuition for Language Model

10/2025

- University of Pennsylvania, Guest lecturer
- BAIR NLP workshop, UC Berkeley, Invited speaker
- University of Washington, NLP seminar, Invited speaker

Bidirectional Language Model

07/2025

- Apple Natural Language Understanding workshop, Invited speaker

Publications

Selected Preprints.....

- [1]: Jie He, Richard He Bai, Sinead Williamson, Jeff Z. Pan, Navdeep Jaitly, **Yizhe Zhang**. CLaRa: Bridging Retrieval and Generation with Continuous Latent Reasoning. [arXiv \(2025\)](#) [*1k GitHub stars*]
- [2]: Haoqiang Kang, **Yizhe Zhang**, Nikki Lijing Kuang, Nicklas Majamaki, Navdeep Jaitly, Yi-An Ma, Lianhui Qin. LaDiR: Latent Diffusion Enhances LLMs for Text Reasoning. [ICLR \(2026\)](#)
- [3]: **Yizhe Zhang**, Haoqiang Kang, Jie He, Navdeep Jaitly. LaDi-RL: Latent Diffusion with Reinforcement Learning for Test-Time Scaling. [arXiv \(2026\)](#) [*20.5 on AIME25 and 52.7 on LCB v6 for 8b model with 2x faster reasoning*]
- [4]: Huangjie Zheng, Shansan Gong, Ruixiang Zhang, Tianrong Chen, Jiatao Gu, Mingyuan Zhou, Navdeep Jaitly, **Yizhe Zhang**. Continuously Augmented Discrete Diffusion Model for Categorical Generative Modeling. [ICLR \(2026\)](#)
- [5]: Amin Karimi Monse, Nikhil Bhendawade, Manuel Rafael Ciosici, Dominic Culver, **Yizhe Zhang**, Irina Belousova. FS-DFM: Fast and Accurate Long Text Generation with Few-Step Diffusion Language Models. [ICLR \(2026\)](#)
- [6]: Shansan Gong, Ruixiang Zhang, Huangjie Zheng, Jiatao Gu, Navdeep Jaitly, Lingpeng Kong, **Yizhe Zhang**. DiffuCoder: Understanding and Improving Masked Diffusion Models for Code Generation. [ICLR \(2026\)](#) [*700 GitHub stars in one month, beats Gemini Diffusion and Qwen coder*]
- [7]: Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, Junteng Liu, Yuntian Deng, **Yizhe Zhang**, Junxian He. Learn to Reason Efficiently with Adaptive Length based Reward Shaping. [ICLR \(2026\)](#)
- [8]: Deepro Choudhury, Sinead Williamson, Adam Goliński, Ning Miao, Freddie Bickford Smith, Michael Kirchhof, **Yizhe Zhang**, Tom Rainforth. BED-LLM: Intelligent Information Gathering with LLMs and Bayesian Experimental Design. [ICLR \(2026\)](#)
- [9]: **Yizhe Zhang**, Richard Bai, Zijin Gu, Ruixiang Zhang, Jiatao Gu, Emmanuel Abbe, Samy Bengio, Navdeep Jaitly. What makes the preferred thinking direction for LLM in Multi-choice Questions? [arXiv \(2025\)](#)
- [10]: Mingyuan Zhou, Yi Gu, Huangjie Zheng, Liangchen Song, Guande He, **Yizhe Zhang**, Wenze Hu, Yinfei Yang. Score Distillation of Flow Matching Models. [arXiv \(2025\)](#)
- [11]: Xiaogeng Liu, Zhiyuan Yu, **Yizhe Zhang**, Ning Zhang, Chaowei Xiao. Automatic and Universal Prompt Injection Attacks Against Large Language Models. [arXiv \(2024\)](#)

Peer-reviewed Conferences and Journals (* equally contributed).....

- [12]: Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, **Yizhe Zhang**, Huangjie Zheng, Tianrong Chen, Miguel Angel Bautista, Josh Susskind, Navdeep Jaitly. Flexible Language Modeling in Continuous

Space with Transformer-based Autoregressive Flows. [NeurIPS \(2025\)](#)

[13]: Jiayi Pan, Xingyao Wang, Graham Neubig, Navdeep Jaitly, Heng Ji, Alane Suhr, **Yizhe Zhang**. Training Software Engineering Agents and Verifiers with SWE-Gym. [ICML \(2025\)](#) [*1.8M+ dataset downloads per month*]

[14]: Ruixiang Zhang, Shuangfei Zhai, **Yizhe Zhang**, James Thornton, Zijing Ou, Joshua Susskind, Navdeep Jaitly. Target Concrete Score Matching: A Holistic Framework for Discrete Diffusion. [ICML \(2025\)](#)

[15]: Yifu Qiu, Varun Embar, **Yizhe Zhang**, Navdeep Jaitly, Shay B. Cohen, Benjamin Han. Eliciting In-context Retrieval and Reasoning for Long-context Large Language Models. [ACL Findings \(2025\)](#)

[16]: Shansan Gong, Shivam Agarwal, **Yizhe Zhang**, Jiacheng Ye, Lin Zheng, Mukai Li, Chenxin An, Peilin Zhao, Wei Bi, Jiawei Han, et al. Scaling Diffusion Language Models via Adaptation from Autoregressive Models. [ICLR \(2025\)](#)

[17]: Xingyao Wang, Boxuan Li, Yufan Song, Frank F. Xu, Xiangru Tang, Mingchen Zhuge, Jiayi Pan, Yueqi Song, Bowen Li, Jaskirat Singh, et al. All-hands: An Open Platform for AI Software Developers as Generalist Agents. [ICLR \(2024\)](#)

[18]: Jiatao Gu, Yuyang Wang, **Yizhe Zhang**, Qihang Zhang, Dinghuai Zhang, Navdeep Jaitly, Josh Susskind, Shuangfei Zhai. DART: Denoising Autoregressive Transformer for Scalable Text-to-Image Generation. [ICLR \(2024\)](#)

[19]: Guoli Yin, Haoping Bai, Shuang Ma, Feng Nan, Yanchao Sun, Zhaoyang Xu, Shen Ma, Jiarui Lu, Xiang Kong, Aonan Zhang, et al. MMAU: A Holistic Benchmark of Agent Capabilities Across Diverse Domains. [NAACL Findings \(2025\)](#)

[20]: Jiarui Lu, Thomas Holleis, **Yizhe Zhang**, Bernhard Aumayer, Feng Nan, Felix Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, et al. Toolsandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use Capabilities. [NAACL Findings \(2025\)](#)

[21]: Jiatao Gu*, Ying Shen*, Shuangfei Zhai, **Yizhe Zhang**, Navdeep Jaitly, Joshua M. Susskind. Kaleido Diffusion: Improving Conditional Diffusion Models with Autoregressive Latent Modeling. [NeurIPS \(2024\)](#)

[22]: Yong Lin, Skyler Seto, Maartje Ter Hoeve, Katherine Metcalf, Barry-John Theobald, Xuan Wang, **Yizhe Zhang**, Chen Huang, Tong Zhang. On the Limited Generalization Capability of the Implicit Reward Model Induced by Direct Preference Optimization. [EMNLP Findings \(2024\)](#)

[23]: Zhuofeng Wu, He Bai, Aonan Zhang, Jiatao Gu, VG Vydiswaran, Navdeep Jaitly, **Yizhe Zhang**. Divide-or-Conquer? Which Part Should You Distill Your LLM? [EMNLP Findings \(2024\)](#)

[24]: **Yizhe Zhang***, He Bai*, Ruixiang Zhang*, Jiatao Gu, Shuangfei Zhai, Josh Susskind, Navdeep Jaitly. How Far Are We from Intelligent Visual Deductive Reasoning? [COLM \(2024\)](#)

[25]: **Yizhe Zhang**, Jiarui Lu, Navdeep Jaitly. The Entity-Deduction Arena: A playground for probing the conversational reasoning and planning capabilities of LLMs. [ACL \(2024\)](#)

[26]: Pratyush Maini, Skyler Seto, He Bai, David Grangier, **Yizhe Zhang**, Navdeep Jaitly. Rephrasing the web: A recipe for compute and data-efficient language modeling. [ACL \(2024\)](#)

[27]: Dinghuai Zhang, **Yizhe Zhang**, Jiatao Gu, Ruixiang Zhang, Josh Susskind, Navdeep Jaitly, Shuangfei Zhai. Improving GFlowNets for Text-to-Image Diffusion Alignment. [ICML SPIGM workshop \(2024\)](#)

[28]: Xingyao Wang, Yangyi Chen, Lifan Yuan, **Yizhe Zhang**, Yunzhu Li, Hao Peng, Heng Ji. Executable code actions elicit better llm agents. [ICML \(2024\)](#)

[29]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Lingjie Liu, Joshua M Susskind. Boot: Data-free distillation of denoising diffusion models with bootstrapping. [ICML \(2024\)](#)

- [30]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Joshua M Susskind, Navdeep Jaitly. Matryoshka diffusion models. [ICLR \(2023\)](#)
- [31]: **Yizhe Zhang**, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Joshua Susskind, Navdeep Jaitly. PLANNER: generating diversified paragraph via latent language diffusion model. [NeurIPS \(2023\)](#)
- [32]: Shuangfei Zhai, Tatiana Likhomanenko, Etai Littwin, Dan Busbridge, Jason Ramapuram, **Yizhe Zhang**, Jiatao Gu, Joshua M Susskind. Stabilizing transformer training by preventing attention entropy collapse. [ICML \(2023\)](#)
- [33]: Felix Faltings, Michel Galley, Baolin Peng, Kianté Brantley, Weixin Cai, **Yizhe Zhang**, Jianfeng Gao, Bill Dolan. Interactive text generation. [EMNLP \(2023\)](#)
- [34]: Jiatao Gu, Shuangfei Zhai, **Yizhe Zhang**, Miguel Angel Bautista, Josh Susskind. f-dm: A multi-stage diffusion model via progressive signal transformation. [ICLR \(2023\)](#)
- [35]: Gyuwan Kim, Jinhyuk Lee, Barlas Oguz, Wenhan Xiong, **Yizhe Zhang**, Yashar Mehdad, William Yang Wang. Bridging the training-inference gap for dense phrase retrieval. [EMNLP Findings \(2022\)](#)
- [36]: **Yizhe Zhang***, Deng Cai*. Linearizing transformer with key-value memory. [EMNLP \(2022\)](#)
- [37]: Tianyu Liu, **Yizhe Zhang**, Chris Brockett, Yi Mao, Zhifang Sui, Weizhu Chen, Bill Dolan. A token-level reference-free hallucination detection benchmark for free-form text generation. [ACL \(2022\)](#)
- [38]: **Yizhe Zhang**, Siqi Sun, Xiang Gao, Yuwei Fang, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Joint retrieval and generation training for grounded text generation. [AAAI \(2022\)](#)
- [39]: Zhisong Zhang, **Yizhe Zhang**, Bill Dolan. Towards More Efficient Insertion Transformer with Fractional Positional Encoding. [EACL \(2021\)](#)
- [40]: Woon Sang Cho, **Yizhe Zhang**, Sudha Rao, Asli Celikyilmaz, Chenyan Xiong, Jianfeng Gao, Mengdi Wang, William B Dolan. Contrastive Multi-document Question Generation. [EACL \(2021\)](#)
- [41]: Jiachang Liu, Dinghan Shen, **Yizhe Zhang**, Bill Dolan, Lawrence Carin, Weizhu Chen. What Makes Good In-Context Examples for GPT-3? [ACL DEELIO workshop \(2021\)](#)
- [42]: Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Bill Dolan. Automatic document sketching: Generating drafts from analogous texts. [ACL Findings \(2021\)](#)
- [43]: Jungo Kasai, Hao Peng, **Yizhe Zhang**, Dani Yogatama, Yi Mao, Weizhu Chen, Noah A Smith. Finetuning Pretrained Transformers into RNNs. [EMNLP \(2021, Oral Presentation\)](#)
- [44]: Dianqi Li, **Yizhe Zhang**, Hao Peng, Liqun Chen, Chris Brockett, Ming-Ting Sun, Bill Dolan. Contextualized perturbation for textual adversarial attack. [NAACL \(2021\)](#)
- [45]: Woon Sang Cho, **Yizhe Zhang**, Sudha Rao, Asli Celikyilmaz, Chenyan Xiong, Jianfeng Gao, Mengdi Wang, Bill Dolan. Unsupervised Common Question Generation from Multiple Documents using Reinforced Contrastive Coordinator. [EACL \(2021\)](#)
- [46]: Zeqiu Wu, Michel Galley, Chris Brockett, **Yizhe Zhang**, Xiang Gao, Chris Quirk, Rik Koncel-Kedziorski, Jianfeng Gao, Hannaneh Hajishirzi, Mari Ostendorf, Bill Dolan. A Controllable Model of Grounded Response Generation. [AAAI \(2021\)](#)
- [47]: Ramakanth Pasunuru, Asli Celikyilmaz, Michel Galley, Chenyan Xiong, **Yizhe Zhang**, Mohit Bansal, Jianfeng Gao. Data Augmentation for Abstractive Query-Focused Multi-Document Summarization. [AAAI \(2021\)](#)
- [48]: **Yizhe Zhang**, Xiang Gao, Sungjin Lee, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Consistent Dialogue Generation with Self-supervised Feature Learning. [SIGDIAL \(2020\)](#)
- [49]: **Yizhe Zhang**, Guoyin Wang, Chunyuan Li, Zhe Gan, Chris Brockett, Bill Dolan. POINTER: Constrained Text Generation via Insertion-based Generative Pre-training. [EMNLP \(2020\)](#)
- [50]: Xiang Gao, **Yizhe Zhang**, Michel Galley, Chris Brockett, Bill Dolan. Dialogue Response Ranking Training with Large-Scale Human Feedback Data. [EMNLP \(2020\)](#)

- [51]: Chunyuan Li, Xiang Gao, Yuan Li, Xiujun Li, Baolin Peng, **Yizhe Zhang**, Jianfeng Gao. Optimus: Organizing Sentences via Pre-trained Modeling of a Latent Space. [EMNLP \(2020\)](#)
- [52]: Jianqiao Li, Chunyuan Li, Guoyin Wang, Hao Fu, Yuhchen Lin, Liqun Chen, **Yizhe Zhang**, Lawrence Carin. Improving Text Generation with Student-Forcing Optimal Transport. [EMNLP \(2020\)](#)
- [53]: Yu Cheng, Zhe Gan, **Yizhe Zhang**, Oussama Elachqar, Dianqi Li, Jingjing Liu. Contextual Text Style Transfer. [Findings of EMNLP \(2020\)](#)
- [54]: Shuyang Dai, Yu Cheng, **Yizhe Zhang**, Zhe Gan, Jingjing Liu, Lawrence Carin. Contrastively Smoothed Class Alignment for Unsupervised Domain Adaptation. [ACCV \(2020\)](#)
- [55]: Xinnuo Xu, **Yizhe Zhang**, Lars Liden, Sungjin Lee. Datasets and Benchmarks for Task-Oriented Log Dialogue Ranking Task. [Interspeech \(2020\)](#)
- [56]: Siyang Yuan, Ke Bai, Liqun Chen, **Yizhe Zhang**, Chenyang Tao, Chunyuan Li, Guoyin Wang, Ricardo Henao, Lawrence Carin. Weakly supervised cross-domain alignment with optimal transport Task. [BMVC \(2020\)](#)
- [57]: **Yizhe Zhang**, Siqi Sun, Michel Galley, Yen-Chun Chen, Chris Brockett, Xiang Gao, Jianfeng Gao, Jingjing Liu, Bill Dolan. DialoGPT: Large-Scale Generative Pre-training for Conversational Response Generation. [demo track, ACL \(2020\)](#) [1,912 citations, 2.4k GitHub stars]
- [58]: Yichen Huang*, **Yizhe Zhang***, Oussama Elachqar, Yu Cheng. INSET: Sentence Infilling with Inter-sentential Generative Pre-training. [ACL \(2020\)](#)
- [59]: Pengyu Cheng, Renqiang Min, Dinghan Shen, Christopher Malon, **Yizhe Zhang**, Yitong Li, Lawrence Carin. Improving Disentangled Text Representation Learning with Information Theoretical Guidance. [ACL \(2020\)](#)
- [60]: Xinjie Fan, **Yizhe Zhang**, Zhendong Wang, Mingyuan Zhou. Adaptive Correlated Monte Carlo for Contextual Categorical Sequence Generation. [ICLR \(2020\)](#)
- [61]: Yuan Li, Chunyuan Li, **Yizhe Zhang**, Xiujun Li, Guoqing Zheng, Lawrence Carin, Jianfeng Gao. Complementary Auxiliary Classifiers for Label-Conditional Text Generation. [AAAI \(2020\)](#)
- [62]: Liqun Chen, Ke Bai, Chenyang Tao, **Yizhe Zhang**, Guoyin Wang, Wenlin Wang, Ricardo Henao, Lawrence Carin. Sequence Generation with Optimal-Transport-Enhanced Reinforcement Learning. [AAAI \(2020\)](#)
- [63]: Xiang Gao, **Yizhe Zhang**, Sungjin Lee, Michel Galley, Chris Brockett, Jianfeng Gao, Bill Dolan. Structuring latent spaces for stylized response generation. [EMNLP \(2019\)](#)
- [64]: Dianqi Li, **Yizhe Zhang**, Zhe Gan, Yu Cheng, Chris Brockett, Ming-Ting Sun, Bill Dolan. Domain Adaptive Text Style Transfer. [EMNLP \(2019\)](#)
- [65]: Xinnuo Xu, **Yizhe Zhang**, Lars Liden, Sungjin Lee. Unsupervised Dialogue Spectrum Generation for Log Dialogue Ranking. [SIGDIAL \(2019\)](#), [Best paper nomination](#)
- [66]: Liqun Chen, Guoyin Wang, Chenyang Tao, Dinghan Shen, **Yizhe Zhang**, Lawrence Carin. Improving Textual Network Embedding with Global Attention via Optimal Transport. [ACL \(2019\)](#)
- [67]: Dinghan Shen, Asli Celikyilmaz, **Yizhe Zhang**, Liqun Chen, Xin Wang, Jianfeng Gao, Lawrence Carin. Towards Generating Long and Coherent Text with Multi-Level Latent Variable Models. [ACL \(2019\)](#)
- [68]: Xiang Gao, Sungjin Lee, **Yizhe Zhang**, Chris Brockett, Michel Galley, Jianfeng Gao, Bill Dolan. Jointly Optimizing Diversity and Relevance in Neural Response Generation. [NAACL \(2019\)](#)
- [69]: Liqun Chen, **Yizhe Zhang**, Ruiyi Zhang, Chenyang Tao, Zhe Gan, Haichao Zhang, Bai Li, Dinghan Shen, Changyou Chen, Lawrence Carin. Improving Sequence-to-Sequence Learning via Optimal Transport. [ICLR \(2019\)](#)
- [70]: **Yizhe Zhang**, Michel Galley, Jianfeng Gao, Zhe Gan, Xiujun Li, Chris Brockett, Bill Dolan. Gen-

- erating Informative and Diverse Conversational Responses via Adversarial Information Maximization. [NeurIPS \(2018\)](#)
- [71]: Liqun Chen, Shuyang Dai, Chenyang Tao, Dinghan Shen, Zhe Gan, Haichao Zhang, **Yizhe Zhang**, Lawrence Carin. Adversarial Text Generation via Feature-Mover’s Distance. [NeurIPS \(2018\)](#)
- [72]: Yunchen Pu, Shuyang Dai, **Yizhe Zhang**, Zhe Gan, Lawrence Carin. Multi-Domain Joint Distribution Learning with Generative Adversarial Nets. [ICML \(2018\)](#)
- [73]: Dinghan Shen, Guoyin Wang, Wenlin Wang, Martin Renqiang Min, Qinliang Su, **Yizhe Zhang**, Chunyuan Li, Ricardo Henao, Lawrence Carin. On Simple Word-Embedding-Based Models and Associated Pooling Mechanisms. [ACL \(2018\)](#)
- [74]: Guoyin Wang, Chunyuan Li, Wenlin Wang, **Yizhe Zhang**, Dinghan Shen, Xinyuan Zhang, Ricardo Henao, Lawrence Carin. Joint Embedding of Words and Labels for Text Classification. [ACL \(2018\)](#)
- [75]: Dinghan Shen, **Yizhe Zhang**, Ricardo Henao, Qinliang Su, Lawrence Carin. Deconvolutional Latent-Variable Model for Text Sequence Matching. [AAAI \(2018\)](#)
- [76]: Wenlin Wang, Piyush Rai, Yunchen Pu, Kai Fan, **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. A Flexible Probabilistic Framework for Learning to Predict Unseen Classes. [AAAI \(2018\)](#)
- [77]: Wenlin Wang, Yunchen Pu, Vinay Verma, Kai Fan, **Yizhe Zhang**, Changyou Chen, Piyush Rai, Lawrence Carin. Zero-shot Learning via Class-Conditioned Deep Generative Models. [AAAI \(2018\)](#)
- [78]: **Yizhe Zhang**, Dinghan Shen, Guoyin Wang, Zhe Gan, Ricardo Henao, Lawrence Carin. Deconvolutional Paragraph Representation Learning. [NIPS \(2017\)](#)
- [79]: Zhe Gan, Liqun Chen, Weiyao Wang, Yunchen Pu, **Yizhe Zhang**, Lawrence Carin. Triangle Generative Adversarial Networks. [NIPS \(2017\)](#)
- [80]: **Yizhe Zhang**, Changyou Chen, Zhe Gan, Lawrence Carin. Stochastic Gradient Monomial Gamma Sampler. [ICML \(2017\)](#)
- [81]: **Yizhe Zhang**, Zhe Gan, Zhi Chen, Lawrence Carin. Adversarial Feature Matching for Text Generation. [ICML \(2017\)](#)
- [82]: **Yizhe Zhang**, Xiangyu Wang, Changyou Chen, Lawrence Carin. Towards Unifying Hamiltonian Monte Carlo and Slice Sampling. [NIPS \(2016\)](#)
- [83]: Changyou Chen, Nan Ding, Chunyuan Li, **Yizhe Zhang**, Lawrence Carin. Distributed Bayesian Learning with Stochastic Gradient MCMC. [NIPS \(2016\)](#)
- [84]: **Yizhe Zhang**, Ricardo Henao, Lawrence Carin. Dynamic Poisson Factor Analysis. [ICDM \(2016\)](#)
- [85]: Kai Fan, **Yizhe Zhang**, Katherine Heller. Triply Stochastic Variational Inference for Non-linear Beta Process Factor Analysis. [ICDM \(2016\)](#)
- [86]: **Yizhe Zhang**, Ricardo Henao, Jianling Zhong, Lawrence Carin, Alexander Hartemink. Learning a Hybrid Architecture for Sequence Regression and Annotation. [AAAI \(2016\)](#)
- [87]: **Yizhe Zhang**, Ricardo Henao, Chunyuan Li, Lawrence Carin. Bayesian Dictionary Learning with Gaussian Processes and Sigmoid Belief Networks. [IJCAI \(2016\)](#)
- [88]: **Yizhe Zhang**, Changyou Chen, Ricardo Henao, Lawrence Carin. Laplacian Hamiltonian Monte Carlo. [ECML \(2016\)](#)
- [89]: **Yizhe Zhang**, Yupeng He, Chaochun Wei. MOST+: a Motif Finding Approach Combining Genomic Sequence and Heterogeneous Genome-wide Signatures. [BMC Genomics \(2015\)](#)
- [90]: Yupeng He, **Yizhe Zhang**, Guangyong Zheng, Chaochun Wei. CRF-based Transcription Factor Binding Site Finding System. [BMC Genomics \(2012\)](#)
- [91]: Jiemeng Liu, Haifeng Wang, Hongxing Yang, **Yizhe Zhang**, Jinfeng Wang, Fangqing Zhao, Ji Qi. Composition-based Classification of Short Metagenomic Sequences Elucidates the Landscapes of Taxonomic and Functional Enrichment of Microorganisms. [Nucleic Acids Research \(2012\)](#)

Workshop, Demo, and Miscellaneous.....

- [92]: Lucie Charlotte Magister, Katherine Metcalf, **Yizhe Zhang**, Maartje Ter Hoeve. On the Way to LLM Personalization: Learning to Remember User Conversations. [ACL L2M2 Workshop \(2025\)](#)
- [93]: **Yizhe Zhang**, Navdeep Jaitly. SAGE: Steering Dialog Generation with Future-Aware State-Action Augmentation. [EMNLP Workshop on NLPerspectives \(2025\)](#)
- [94]: Tatiana Likhomanenko, Luke Carlson, Richard He Bai, Zijin Gu, Han Tran, Zakaria Aldeneh, **Yizhe Zhang**, Ruixiang Zhang, Huangjie Zheng, Navdeep Jaitly. ChipChat: Low-Latency Cascaded Conversational Agent in MLX. [ASRU demo track \(2025\)](#)
- [95]: Shangshang Zheng, He Bai, **Yizhe Zhang**, Yi Su, Xiaochuan Niu, Navdeep Jaitly. KGLens: Towards Efficient and Effective Knowledge Probing of Large Language Models with Knowledge Graphs. [ACL Towards Knowledgeable Language Models Workshop \(2024\)](#)
- [96]: Georgia Gabriela Sampaio, Ruixiang Zhang, Shuangfei Zhai, Jiatao Gu, Josh Susskind, Navdeep Jaitly, **Yizhe Zhang**. TypeScore: A Text Fidelity Metric for Text-to-Image Generative Models. [arXiv \(2024\)](#)
- [97]: Ying Shen, **Yizhe Zhang**, Shuangfei Zhai, Lifu Huang, Joshua M. Susskind, Jiatao Gu. Many-to-Many Image Generation with Auto-Regressive Diffusion Models. [ICML 2024 Workshop on Structured Probabilistic Inference Generative Modeling \(2024\)](#)
- [98]: Shikib Mehri, Jinho Choi, Luis Fernando D'Haro, Jan Deriu, Maxine Eskenazi, Milica Gasic, Kallirroi Georgila, Dilek Hakkani-Tur, Zekang Li, Verena Rieser, et al. Report from the NSF Future Directions Workshop on Automatic Evaluation of Dialog: Research Directions and Challenges. [NSF AED Workshop \(2022\)](#)
- [99]: Vighnesh Leonardo Shiv, Chris Quirk, Anshuman Suri, Xiang Gao, Khuram Shahid, Nithya Govindarajan, **Yizhe Zhang**, Jianfeng Gao, Michel Galley, Chris Brockett. Microsoft Icecaps: An Open-Source Toolkit for Conversation Modeling. [ACL System Demonstrations \(2019\)](#)
- [100]: Woon Sang Cho, Pengchuan Zhang, **Yizhe Zhang**, Xiujun Li, Michel Galley, Chris Brockett, Mengdi Wang, Jianfeng Gao. Towards Coherent and Cohesive Long-form Text Generation. [Workshop on Narrative Understanding \(2019\)](#)

Teaching

Duke: Advanced Machine Learning (STA571), Instructor: Katherine Heller

Duke: Probabilistic Machine Learning (CS571), Instructor: Cynthia Rudin

Honors and Awards

2023–Present: Stanford top 2% scientists

2018: NeurIPS top 5% reviewer award

2008–2011: Department Fellowship

2012: National Excellent Graduate Scholarship (top 1%)

Travel Awards: NIPS (2015, 2016), ICML (2017), ICDM (2016), IJCAI (2016), AAAI (2016)

Technical Proficiency

Programming: PyTorch, TensorFlow, C/C++, Python, Java, Lua, MATLAB, R